SAMPLE

Great Observatories: LOFAR

The Low-Frequency Array

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Introduction

LOFAR, the LOw-Frequency ARray, is a radio telescope unlike any that came before it: not a dish, not a collection of dishes, but a forest of simple antennas, tens of thousands of them, spread across a swathe of northern Europe, wired together and pointed not by machinery but by mathematics. Where the great dishes of the century's radio astronomy gathered light like basins gather rain, LOFAR listens with the electronic equivalent of an insect's eye, and it points its gaze by computing, forming beams out of raw antenna signals and steering them across the sky at the speed of software.

The array was built by ASTRON, the Netherlands Institute for Radio Astronomy, opening in 2010, and its logic answered a question the field had circled for decades. Below roughly 250 megahertz, the longest wavelengths that penetrate the atmosphere, the sky is a continent no telescope had ever mapped properly: the dishes grew distorted at those wavelengths, their surfaces intolerant of waves measured in meters, and the ionosphere, turbulent at those frequencies, smeared the images as weather smears a photograph. LOFAR's answer was to give up the dish entirely and to fight the ionosphere with computing, replacing the century's most expensive metal with the century's cheapest electronics and the hardest software.

The array's geography is part of its wonder. The core, a dense cluster of stations near Exloo in the Dutch province of Drenthe, sits on a raised mound the builders call the superterp, and from there the stations run outward: across the Netherlands in the first few hundred kilometers, then across borders, to Germany, Poland, France, Sweden, the UK, Ireland, Italy, Latvia, and beyond, the farthest stations separated by roughly two thousand kilometers, a telescope whose collecting field spans a continent and whose resolving power, at the longest baselines, ranks with the finest instruments on Earth.

What the array is for, the chapters ahead set out in order: the astronomy of the lowest frequencies, which the dishes never mastered; the epoch of reionization, the cosmic dawn when the first stars switched on, whose hydrogen signal arrives stretched to the very band LOFAR hears; the surveys, LoTSS among them, that have cataloged more radio sources at these wavelengths than all previous low-frequency maps combined; the Sun, the cosmic rays, the pulsars, and the transients; and the record-holding discoveries, the giant radio galaxies, that the deep maps keep surfacing. One instrument, no moving parts, a continent wide, reading the sky's oldest light, and this volume is its account.

The name unpacks in two words, and the unpacking is the whole design: low frequency, meaning the long radio wavelengths the atmosphere transmits but the century's dishes neglected, and array, meaning that the telescope is not one instrument but a consensus of thousands of antennas, each worthless alone, all of them together a telescope. The reader who holds the two words holds the chapters' subject: an instrument built for the sky's oldest band, organized as a network rather than an object, and the book's account moves from the physics of that choice to the consequences that have followed it, the surveys, the records, and the collaborations.

A note on the volume's own scale of reference, offered at the start so the later chapters can move quickly. The frequencies in this book run from a few tens to a few hundred megahertz; the wavelengths from tens of meters down to roughly one; the array's stations from the dense Dutch core to the outposts at two thousand kilometers; and the science from particles striking the atmosphere overhead to hydrogen shaking itself into the first stars ten-something billion years ago. The range is the point: nothing else in the astronomer's toolkit spans so wide a set of scales in a single instrument, and the chapters will ask the reader to move between the scales freely, because the array does.

Two metaphors organize this account, and naming them at the outset will save the chapters repetition. The first is the ear: the array's dipoles, individually deaf, collectively discerning, recall the biological arrays of hearing, and the volume's physics chapters will return to the resemblance, because the array's images are built the way a listener locates a sound, from the differences of arrival among many receivers. The second is the network: the telescope is a collaboration in hardware, stations across a continent agreeing by fiber and clock to be one instrument, and the volume's institutional chapters will treat that agreement as the design's heart rather than its bureaucracy. Ear and network, listening and agreeing, are the array's two natures, and every discovery in the pages ahead is both at once, a consensus of deaf ears, which is also a fair description of science itself.

Chapter 1: The Discovery of LOFAR

The conception of LOFAR began with a Dutch frustration, and the discovery story of the telescope is the story of

that frustration turned into a plan. Radio astronomy was born in the low frequencies: Jansky's 1932 hiss was at 20 megahertz, the early Australian surveys mapped the sky at meter wavelengths, and the field's first maps, the southern sky's and Reber's northern contours, were made in the band that the later, larger dishes would abandon. As the dishes grew, the profession climbed in frequency, chasing resolution and freedom from ionospheric noise, and by the end of the twentieth century the lowest frequencies, below about 250 megahertz, were the field's abandoned ground floor: known to be rich, known to be hard, effectively unobserved.

The plan that became LOFAR answered with a reversal. Instead of fighting the long wavelengths with larger, more precise surfaces, the Dutch engineers proposed to give up the surface altogether: an antenna need only be a simple dipole, a metal stick a meter or two across, cheap enough to deploy by the tens of thousands, and the telescope's directivity, its pointing and its resolving power, could come instead from the timing of signals across a vast array. The proposal, developed at ASTRON through the early 2000s and adopted into the Dutch national roadmap, was radical in its economy: a telescope with no precision manufacturing at all, whose entire cost lived in its computing.

The array grew fast once the decision was made, and the dates tell it plainly. Prototype stations and the first observing campaigns ran in the middle years of the decade; the core took shape around Exloo through 2007 to 2009, the stations spreading across the Netherlands on the same national fiber that had been laid for other purposes; and in June 2010 the telescope was officially opened, already the largest connected radio instrument in the world for its band. From the first, the design reached past the border: partner institutions in Germany and Poland joined with stations of their own, and the array's baselines, hundreds of kilometers, stretched toward the thousands as France, Sweden, the United Kingdom, Ireland, Italy, and Latvia added their stations in the years after, each antenna field extending the telescope by a country.

The discovery chapter closes on the concept's vindication, because the reversal worked. The simple antennas heard the low sky, the software pointed the beams, and the ionosphere, the ancient enemy, proved calibratable at the precision the arrays of software demanded. Within its first decade LOFAR was producing the deepest low-frequency maps ever made, the first catalogs of the meter-wave sky, and the first detections of phenomena, from the deepest radio galaxies to the shower fronts of cosmic rays, that the abandoned ground floor had been hiding for the whole history of the field, and the instrument that many had dismissed as a gamble is now the template for the low-frequency half of the coming radio astronomy.

The conception's technical ancestry merits a paragraph, because the array's ideas were not born new. Phased arrays had served radar since the mid-century; radio interferometry's mathematics had been working since Ryle's groups and the Dutch themselves had refined it at Westerbork; and the low-frequency sky had been observed, coarsely, by the Cambridge catalogs and their like at 178 and 81 megahertz, the famous 3C and 4C lists whose sources founded quasar science. What the Dutch plan added was the convergence: dipole antennas cheap enough for tens of thousands, fiber networks to bind them, digital receivers on every element, and computing power to do what the ionosphere had always prevented, and the addition was an engineering economy rather than a physics discovery, which is the kind of progress the telescopes' history quietly runs on.

The construction story carries a second act worth the telling: the array's growth into an international instrument began almost before the core was finished. German institutes, through the first decade of the new century, committed to stations at their own sites, Effelsberg's neighborhood among them; Poland followed with three stations; and the pattern repeated, station by station, as partners from France, Ireland, Sweden, Italy, Latvia, and Britain joined in the years that followed, each with the same hardware and the same fiber obligation, so that the telescope's borders advanced by accession rather than conquest, a scientific common market with antennas as its currency.

The conception's institutional driver merits the chapter's last paragraph, because the array was born inside ASTRON's particular culture. The institute, the descendant of the Dutch radio laboratories of the 1940s and 1950s, had spent the post-war decades building an unbroken line of instruments, the Dwingeloo dish, the Westerbork array, and the technical groups that had pioneered receiver and correlator design for the world's arrays, and the LOFAR plan was that culture's answer to a changing era: when the traditional dish's costs began to scale against the possibilities of digital electronics, the institute chose the electronics, and the choice was made with the confidence of a house that had been right about instruments before, a confidence the partner nations, joining station by station through the decade, quietly endorsed with their funding.

The early milestone sequence deserves the story's last paragraph, because the array's first years read as a public proof of every promise. The first survey, the multifrequency snapshot program that mapped the northern sky at the

modest depths of the commissioning era, showed that a dipole telescope could survey at all; the first deep images of the HBA test fields showed that the ionosphere could be calibrated to the arcsecond class; the first pulsar detections at the new sensitivity, and the first solar interferograms, showed the tied-beam modes working; and by the middle of the decade the Two-metre Sky Survey's first data release, with its hundreds of thousands of sources, closed the argument, the telescope's promises kept in print, milestone by milestone, within five years of the opening, a pace the classical instruments rarely matched and the software instruments now promise by default.

One more conception story belongs in the chapter, because the array's design lineage passes through a cancelled dream. The Dutch community, in the decades before LOFAR, had drawn plans for a next-generation square-kilometer collecting area, the international project that would become the Square Kilometre Array, and the long planning of that instrument taught the community its economics: that the classical dish's cost curve could not deliver the collecting area the science wanted, and that the computing curve, falling year by year, might. When the SKA's own timeline stretched into the century's third decade, the Dutch institutes turned the lesson into an instrument they could build at once, and the array's design is therefore the SKA's argument, made early, in hardware, at the low band, a proving ground the larger project now cites, which is the chapter's closing irony: the telescope that began as the rehearsal has become, for its band, the performance.